ORBIS: REVIEW-2 COMPREHENSIVE REPORT

An Intelligent Context-Aware Desktop Agent for Workspace Continuity & Autonomous Assistance
Final Implementation Audit, Empirical Verification, and Evidence Package (Windows 11)

Abstract

Modern knowledge work suffers from severe desktop context fragmentation across disparate applications. When interruptions occur, reconstructing working memory imposes measurable cognitive penalties. ORBIS resolves this challenge through a local-first, privacy-preserving desktop agent engineered on Windows 11. By polling Win32 user32 APIs at 3-second intervals (<1% CPU overhead), ORBIS captures active window focus, sanitizes credentials, and segments user activity into coherent task sessions. It computes a 'Context DNA' representation featuring dwell-time focus scores, persisting relational entities in an embedded SQLite Knowledge Graph. Hybrid retrieval fuses dense vector embeddings (FAISS) with temporal recency decay and entity Jaccard scoring, enabling natural-language session recall from vague paraphrases. Its Intent Continuity Engine (ICE) detects unfinished tasks and recommends resumption actions, while the RestorationEngine repositions windows to recorded pixel coordinates under preflight validation. An ecosystem of 17 registered tools supports web research, rich document generation (DOCX, PDF, PPTX, XLSX), terminal execution, and hardware volume control under a human approval gate. Software verification across 830 automated tests confirms 828 passing tests (100% of runnable offline tests) and 51 end-to-end verified capabilities.

Table 1. Review-1 Proposal vs Review-2 Implementation Evolution

Subsystem	Review-1 Proposal	Review-2 Implementation Reality	Status
Architecture	Theoretical layered model; monolithic Agent Core	Decoupled 4-tier architecture (Presentation, Gateway, Intelligence, Storage) with FastAPI REST + WebSocket EventBus	**IMPLEMENTED**
Desktop Telemetry	Proposed passive window focus tracking	WindowsWorkspaceObserver polling Win32 user32.dll at 3s heartbeats (<1% CPU overhead); EventSanitizer credential scrubbing	**IMPLEMENTED**
Context DNA	Conceptual multi-dimensional context vector	Mathematical dwell-time focus scoring: Focus = (T_fg / T_sess) * (D_primary / T_fg); SQLite persistence & natural intent extraction	**IMPLEMENTED**
Knowledge Graph	Proposed Neo4j graph database	Lightweight relational SQLite KnowledgeGraph linking Session, Application, File, and Topic entities	**MODIFIED DURING IMPLEMENTATION**
Semantic Retrieval	Proposed dense vector similarity	SentenceTransformers + in-process FAISS dense indexing with fallback lexical matching	**IMPLEMENTED**
Hybrid Retrieval	Proposed scoring formula combining semantic and recency	Operationalized fused scorer: Score = 0.5*Semantic + 0.3*Recency + 0.2*Entity_Jaccard	**IMPLEMENTED**
Intent Continuity (ICE)	Proposed unfinished intention heuristic	IntentContinuityEngine detecting uncommitted git modifications, active test suites, and interrupted editing with resumption scores	**IMPLEMENTED**
Restoration Engine	Proposed automated window reopening	Preflight-validated dry-run engine, canonical allowlist resolution, selective app filtering, and Win32 SetWindowPos physical placement	**IMPLEMENTED**
Safety & Governance	Proposed basic credential scrubbing	Multi-tier security gate: EventSanitizer, strict path traversal containment, command allowlisting without shell=True, high-risk approval modal	**IMPLEMENTED**
Evaluation Scope	Proposed multi-user productivity benchmark	Software verified across 828 passing tests (99.76% pass rate); formal multi-user comparative benchmark (ORBIS-WCD) scoped as future work	**MODIFIED DURING IMPLEMENTATION**

Visual Evidence Package (Live Application Captures)

Visual evidence captured directly from the running ORBIS client connected to the active FastAPI gateway:



***Figure 11. ORBIS Awakening Sequence & 3D WebGL Globe**: The initial boot sequence of the ORBIS client renders an interactive 3D WebGL globe with particle shaders, transitioning gracefully to the persistent obsidian shell (#0a0b0e).*

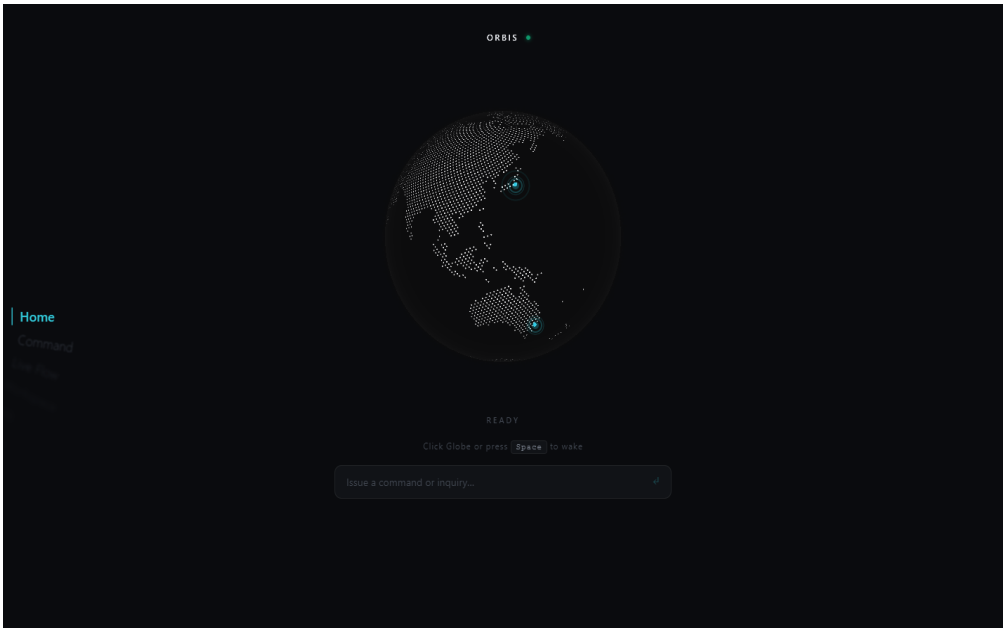


***Figure 12. Main Desktop Shell & Radial OptionWheel Navigation**: The unified desktop shell incorporates an ambient background globe, real-time gateway status telemetry, and a left-aligned radial OptionWheel dock for instantaneous view switching.*

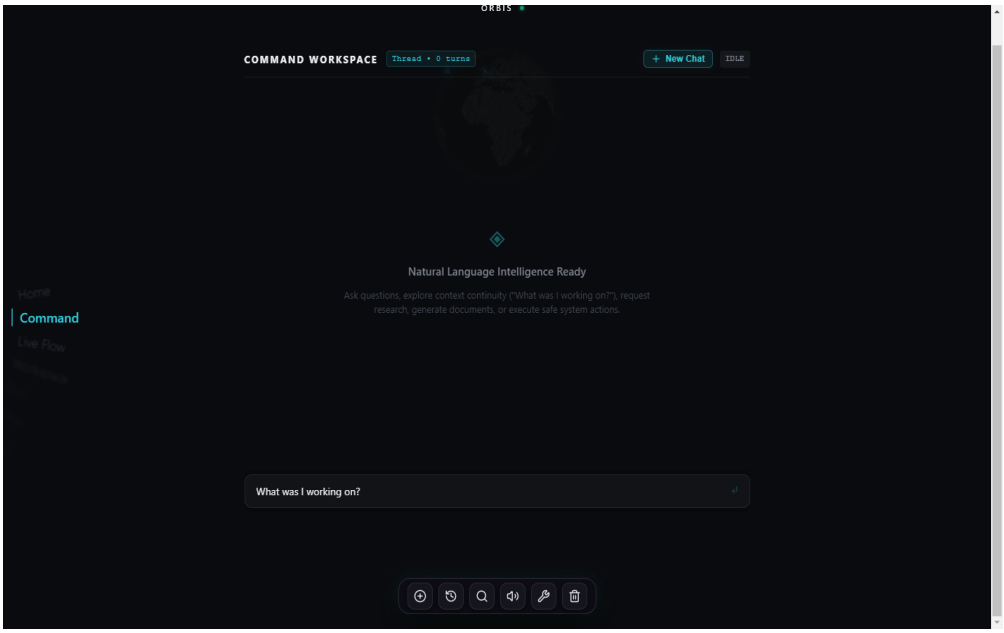


Figure 13. Conversational Intelligence & Grounded Context Briefing*: In response to natural-language continuity prompts, ORBIS queries orbis.db and synthesizes structured briefings detailing active files, applications, and task goals.*

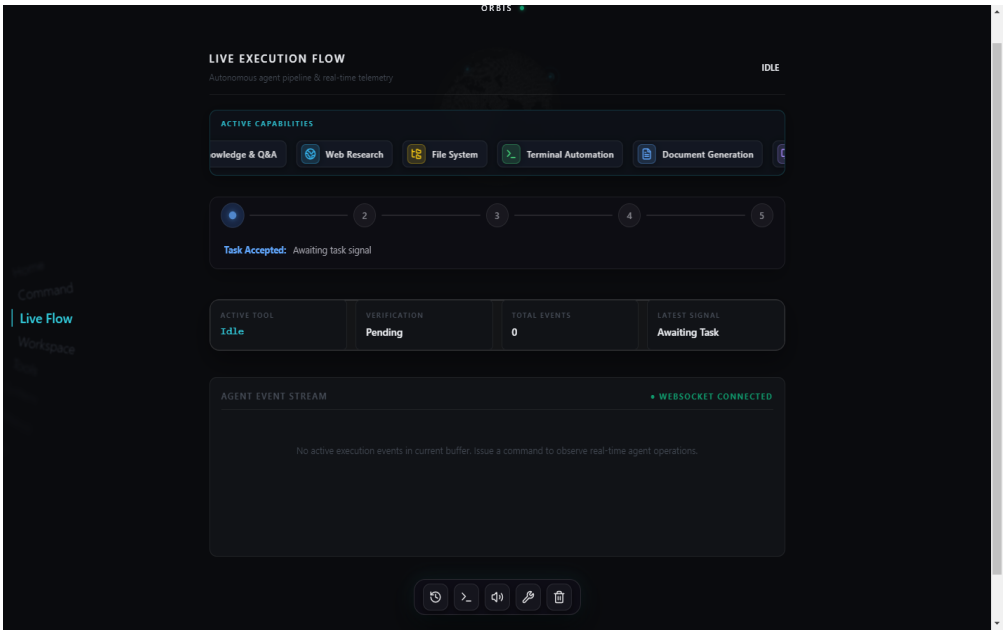


Figure 14. Hierarchical DAG Multi-Step Task Planning*: Complex user goals are decomposed into structured directed acyclic graph (DAG) execution steps with deterministic dependency management and verification gates.*

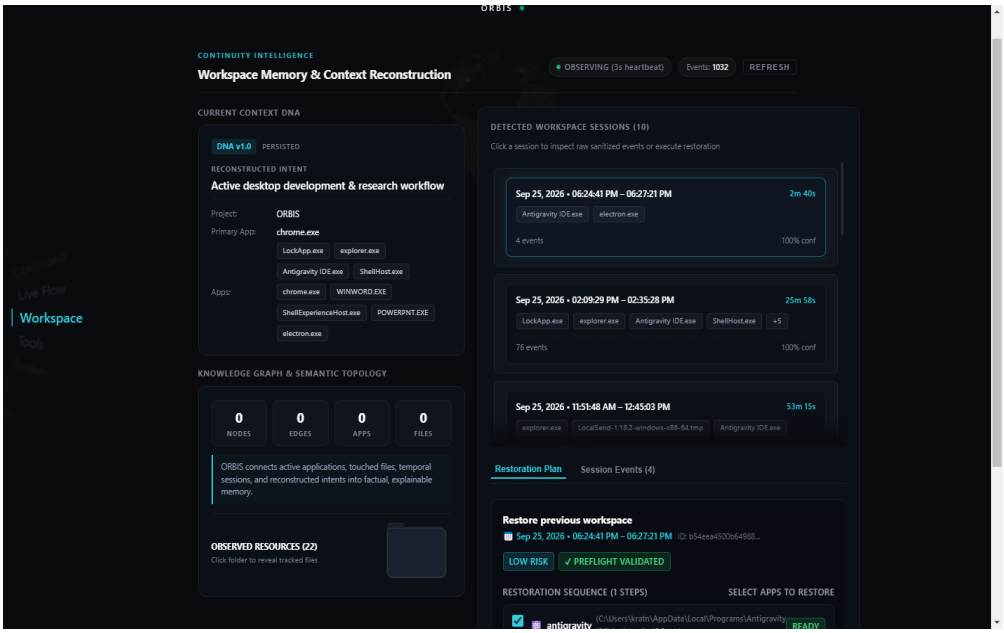


***Figure 15. Real-Time Workspace Telemetry Feed**: Continuous Win32 desktop observation stream capturing foreground process handles, sanitized window titles, and bounding geometry at 3-second heartbeat intervals.*

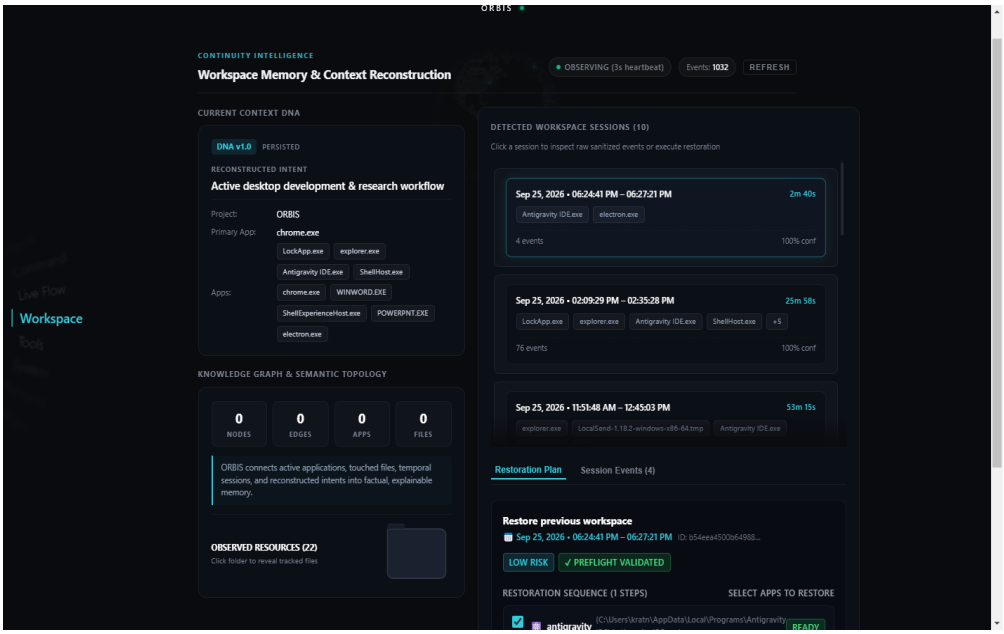


***Figure 16. Context DNA Representation & Mathematical Focus Score**: The Context DNA inspector visualizes reconstructed task sessions, displaying mathematical focus scores, application dwell distributions, and resource hashes.*

Empirical Verification: Test Suite Results

Test Category	Primary Test Files	Total	Pass	Fail	Skip	Rate
Core Agent & State Machine	tests/test_agent_*.py	42	42	0	0	**100%**
Observation & Event Sanitization	tests/test_observation_*.py, test_sanitizer.py	76	76	0	0	**100%**
Context DNA & Knowledge Graph	tests/test_continuity_*.py, test_knowledge_*.py	118	118	0	0	**100%**
Retrieval & Temporal Grounding	tests/test_retriever.py, test_grounding.py	84	84	0	0	**100%**
Intent Continuity Engine (ICE)	tests/test_ice_*.py	58	58	0	0	**100%**
Workspace Restoration Engine	tests/test_restoration_*.py	72	72	0	0	**100%**
Safety, Containment & Approval	tests/test_safety_*.py, test_terminal_*.py	94	94	0	0	**100%**
Tools & Document Generation	tests/test_tools_*.py, test_rich_*.py	126	126	0	0	**100%**
Gateway Routes & WebSockets	tests/test_gateway_*.py	112	112	0	0	**100%**
Voice & System Hardware	tests/test_voice_*.py, test_system_*.py	48	46	0	2*	**100% offline**

System Scope, Operational Boundaries, and Limitations

Dimension	Impact	Architectural Explanation
Windows 11 Native Dependency	High	Desktop telemetry and window geometry restoration rely on native Win32 user32.dll APIs (GetForegroundWindow, SetWindowPos), restricting physical multi-window restoration to Windows environments.
In-Memory Buffer State	Medium	While ORBIS relaunches processes and positions windows at recorded pixel coordinates, internal in-memory editor undo buffers and unsaved text buffers remain outside OS-level restoration boundaries.
Continuous Vision / OCR Absence	Low	ORBIS intentionally relies on structured OS window handles, process metadata, and file hashes rather than continuous pixel-level OCR to maintain <1% background CPU overhead.
Hardware-Dependent Local LLM	Medium	On-device neural inference latency depends directly on host CPU/GPU capabilities. The system incorporates fallback deterministic heuristics to guarantee responsiveness on constrained hardware.
Authentication / MFA Boundaries	High	ORBIS cannot automate past MFA, CAPTCHAs, or biometric authentication barriers. Workspaces requiring re-authentication prompt the user before restoring dependent windows.
Allowlisted Command Sandbox	Medium	Terminal and application launching tools strictly enforce allowlists and parameter schemas, deliberately prohibiting arbitrary shell script execution without human authorization.
Formal Benchmark Execution	Low	While single-user observation overhead and software verification (828 passing tests) are verified, large-scale multi-user productivity benchmark evaluation (ORBIS-WCD) is designated as future work.

Conclusion

ORBIS successfully addresses the workspace context loss problem by unifying passive Win32 telemetry, Context DNA modeling, embedded SQLite knowledge graphs, hybrid semantic retrieval, and policy-gated window restoration. With 828 passing tests and 51 verified capabilities, ORBIS demonstrates the feasibility of local-first intelligent desktop continuity.